

ReceiptIQ - Server Guide

Project server par kaise chalta hai - Roman Urdu

1. Sab se pehle: app ko chalana nahi parta

ReceiptIQ server par **systemd** ke tahat chalti hai. Iska matlab yeh hai ke woh khud start hoti hai, crash ho jaye to khud wapas aa jati hai, aur server reboot ho to bhi khud chal padti hai.

Aapko koi command chalane ki zaroorat nahi. SSH band kar dein, laptop band kar dein, OpenVPN disconnect kar dein - app chalti rahegi. Woh server par hai, aapke laptop par nahi.

2. Server par kya kya chal raha hai

Service	Kaam
receiptiq	Django app, gunicorn ke zariye, port 8001 par
receiptiq-ngrok	VPN ke baghair khulne wala raasta
receiptiq-tunnel	Cloudflare ka backup raasta
postgresql	Database (receiptiq_db)
ollama	AI models, sirf localhost:11434 par - bahar se band

3. Kaun sa link kab chalta hai

OpenVPN	Address
Connected	http://10.0.2.13:8001
Disconnected	https://unaroused-uncontributively-kalyn.ngrok-free.dev
Disconnected	https://earth-prediction-hostels-optimization.trycloudflare.com (backup)
Disconnected	http://10.0.2.13:8001 - NAHI chalega

Wajah: 10.0.2.13 ek private address hai. Woh sirf VPN ke andar maani rakhta hai. Internet par aisa koi pata maujood nahi hota, is liye VPN band hote hi raasta khatam ho jata hai. App phir bhi chal rahi hoti hai - sirf us tak pohanchne ka raasta badalna parta hai.

4. Settings kahan hain

Port aur tunnel ka link project ki apni **.env** file mein hain:

```
~/receiptiq/.env

APP_PORT=8001
GUNICORN_WORKERS=3
GUNICORN_TIMEOUT=600
NGROK_URL=https://unaroused-uncontributively-kalyn.ngrok-free.dev
```

Har service ek chhoti script chalati hai jo .env parh kar values uthati hai. Service files mein koi port ya link likha hua nahi:

```
~/receiptiq/deploy/run-app.sh -> gunicorn, APP_PORT par
~/receiptiq/deploy/run-ngrok.sh -> ngrok, NGROK_URL par
~/receiptiq/deploy/run-cloudflared.sh -> Cloudflare tunnel
```

```
/etc/systemd/system/receiptiq.service
ExecStart=/home/siresoft/receiptiq/deploy/run-app.sh
```

Port badalna ho to sirf .env mein badlein, phir services restart kar dein. Teenon khud naya port utha lengi.

```
sudo systemctl restart receiptiq receiptiq-ngrok receiptiq-tunnel
```

Note: yeh scripts is liye zaroori hain ke systemd khud .env nahi parh sakta - SELinux use home folder se file parhne nahi deta. Script 'siresoft' ke tor par chalti hai, jiske paas us file ki ijazat hai.

Note: Django ki settings.py mein port kabhi nahi hota. Django ko pata hi nahi hota ke woh kis port par chal rahi hai - port gunicorn tay karta hai.

5. Roz ke commands

Server par jana:

```
ssh siresoft@10.0.2.13
cd ~/receiptiq
```

Sab theek hai ya nahi (99% waqt sirf yeh):

```
systemctl is-active receiptiq receiptiq-ngrok receiptiq-tunnel
```

Teen lines aayengi - teenon 'active' honi chahiye.

App jawab de rahi hai ya nahi:

```
curl -I http://10.0.2.13:8001/
```

HTTP 302 ya 200 = theek hai.

Code ya .env badalne ke baad:

```
sudo systemctl restart receiptiq
```

Logs dekhne ke liye:

```
journalctl -u receiptiq -n 50 --no-pager # aakhri 50 lines
journalctl -u receiptiq -f # live (Ctrl+C se niklein)
journalctl -u receiptiq -p err -n 30 --no-pager # sirf errors
```

6. Django ke kaam

```
cd ~/receiptiq
source venv/bin/activate

python manage.py migrate
python manage.py createsuperuser
python manage.py collectstatic --noinput

deactivate
sudo systemctl restart receiptiq
```

Code test karna ho to hamesha alag port par - 8001 asli app ne le rakha hai:

```
python manage.py runserver 0.0.0.0:8002
```

7. Ehtiyat

Command	Kya hoga
sudo systemctl stop receiptiq	App band ho jayegi
sudo systemctl stop receiptiq-ngrok	VPN-free raasta band
sudo systemctl disable receiptiq	Reboot par khud nahi chalegi
rm	File hamesha ke liye khatam
cat > file	Purana content foran mit jata hai

Agar screen ruk jaye aur neeche ':' ya '(END)' nazar aaye - q dabayein. Agar terminal '>' par atak jaye - **Ctrl+C** dabayein.

8. Jo abhi baqi hai

Maujooda public links (ngrok aur Cloudflare) muft tunnel addresses hain. Testing aur demo ke liye theek hain, magar company ke asli product ka pata yeh nahi hona chahiye.

https://receipts.siresoft.com jaisa sthir naam banane ke liye do mein se ek cheez chahiye - aur dono network/DNS ka kaam hain, app ka nahi:

1. Koi ek siresoft domain muft Cloudflare account par lana. Uske baad Cloudflare Tunnel khud hostname aur HTTPS de deta hai, bilkul muft. Taqreeban 10 minute ka kaam.
2. Ya router (10.0.2.1) par port forward: external TCP 8001 -> 10.0.2.13:8001, saath mein DNS record aur server par nginx + SSL certificate.

ngrok bhi aazmaya gaya - woh apne chune hue naam ke paise leta hai, is liye munasib nahi.

siresoft.com abhi apne nameservers (ns1/ns2.siresoft.com) par hai, is liye yeh kaam us shakhs ka hai jiske paas DNS ka access ho.